

1 True or False

1. Hosts usually perform the iterative DNS resolution process themselves.
2. Every zone always has at least 2 name servers.
3. When looking up a root server, BGP will use unicast to find the correct root server.
4. A client can establish a TCP connection with a root server.
5. Most queries to DNS root servers are for nonexistent TLDs.

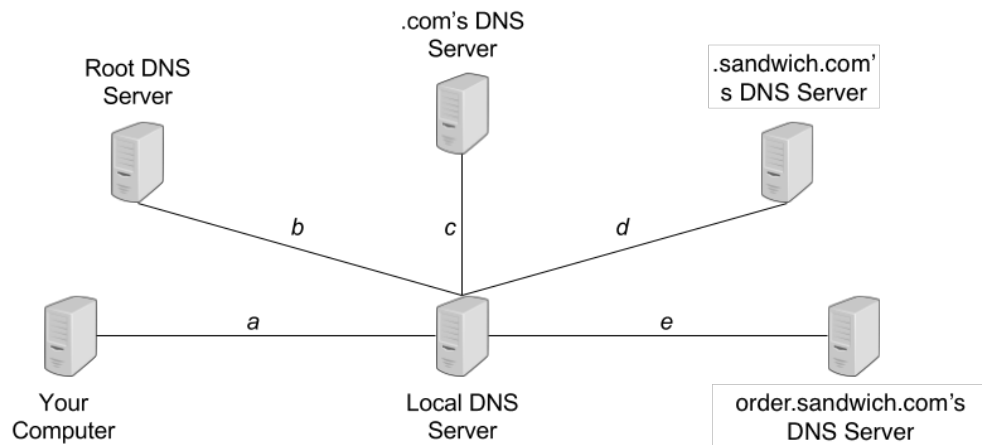
2 DNS Record Types

Write the corresponding DNS record type on the right column.

Information	Record Type
Name to IPV4 Address Mapping	
Name to IPV6 Address Mapping	
Name Server	
Human Readable Information (Often Used for Site Verification)	
General Name-to-Service Mapping	
Mail Exchanger	
Canonical Name	

3 DNS

A sandwich ordering website `www.order.sandwich.com` is accepting online orders for the next T minutes. Consider the following setup of DNS servers, with annotated latencies between servers:



Assume that the latency between your computer and the website's server is t , that once you send an order for a sandwich you must wait for a confirmation response from the website before issuing another, and that your computer does not cache website's IP address.

Meta: If students ask whether these DNS servers use iterative or recursive DNS lookups, ask which would be more performant for major DNS servers with heavy traffic. (This was covered in lecture.)

In each of the following three scenarios below, determine how many sandwiches you can order for the next T minutes:

- (1) Your local DNS server doesn't cache any information.
- (2) Your local DNS server caches responses, with a time-to-live $L \geq T$.
- (3) Let $T = 600$ seconds and $a = b = c = d = e = t = 1$ second. Your local DNS server caches responses with a finite time-to-live of 30 seconds.